Submission Date : 9/3/2026

Write a C++ program for a Vehicle Registration System using inheritance, runtime polymorphism, templates, and file handling.

1) Abstract Base Class: Vehicle

Properties: RegistrationNumber, ModelName, Manufacturer, Year.

Pure virtual function: displayInfo().

Validation: Year should be between 2000 and the current year.

2) Derived Classes

Car

Adds property: FuelType

Bike

Adds property: EngineCapacity

3) Other Requirements

Use runtime polymorphism (Vehicle* or smart pointers) to manage both Car and Bike objects.

Use a template function to validate the input year.

Use a static data member in Vehicle to count how many vehicle records are created.

Use an inline function to format and return the vehicle display string.

Store vehicle objects into a binary file, then read them back and display the records.

Implement exception handling for invalid inputs (e.g., empty registration number, invalid year).

Sample Input

Enter number of vehicles: 2

Enter type of vehicle (C for Car, B for Bike): C

Enter Registration Number: MH20AB1234

Enter Model Name: Swift

Enter Manufacturer: Maruti

Enter Year: 2018

Enter Fuel Type: Petrol

Enter type of vehicle (C for Car, B for Bike): B

Enter Registration Number: MH20XY5678

Enter Model Name: Pulsar

Enter Manufacturer: Bajaj

Enter Year: 2020

Enter Engine Capacity: 150cc

Expected Output (after reading from file)

RegNo: MH20AB1234, Model: Swift, Manufacturer: Maruti, Year: 2018, FuelType: Petrol

RegNo: MH20XY5678, Model: Pulsar, Manufacturer: Bajaj, Year: 2020, EngineCapacity: 150cc

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Total Vehicles stored: 2